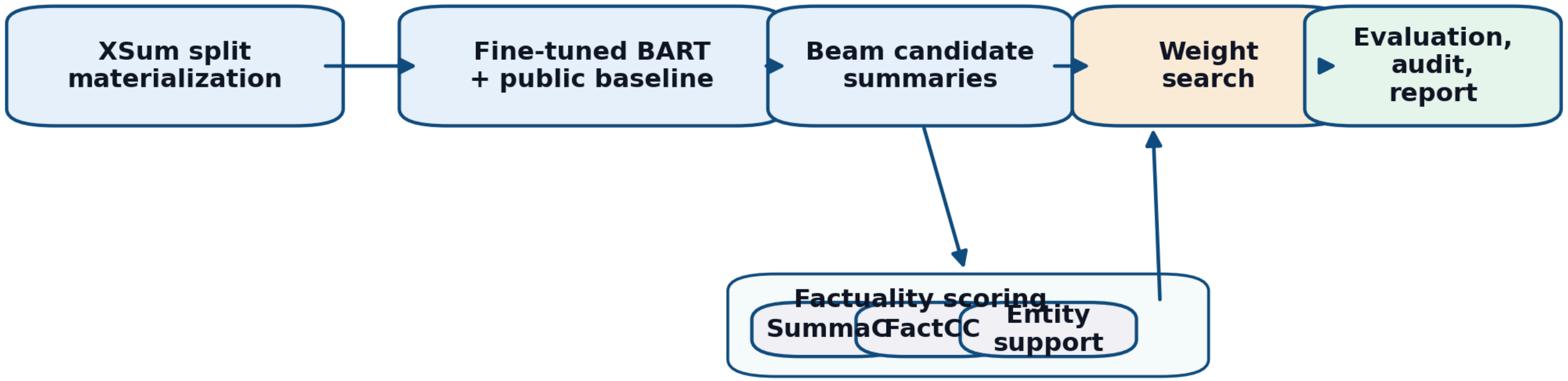
Problem and Setup

- XSum rewards compression, so fluent hallucinations are common.
- Overlap metrics alone do not tell us whether a summary is supported by the article.
- Keep the public `facebook/bart-large-xsum` path as the baseline.
- Headline trade-off: factuality composite `0.3324 -> 0.4420`, ROUGE-Lsum `0.3569 -> 0.3398`
- Bounded scope: `train=128`, `val_tune=64`, `val_full=128`, `test=128`, qualitative subset `=24`

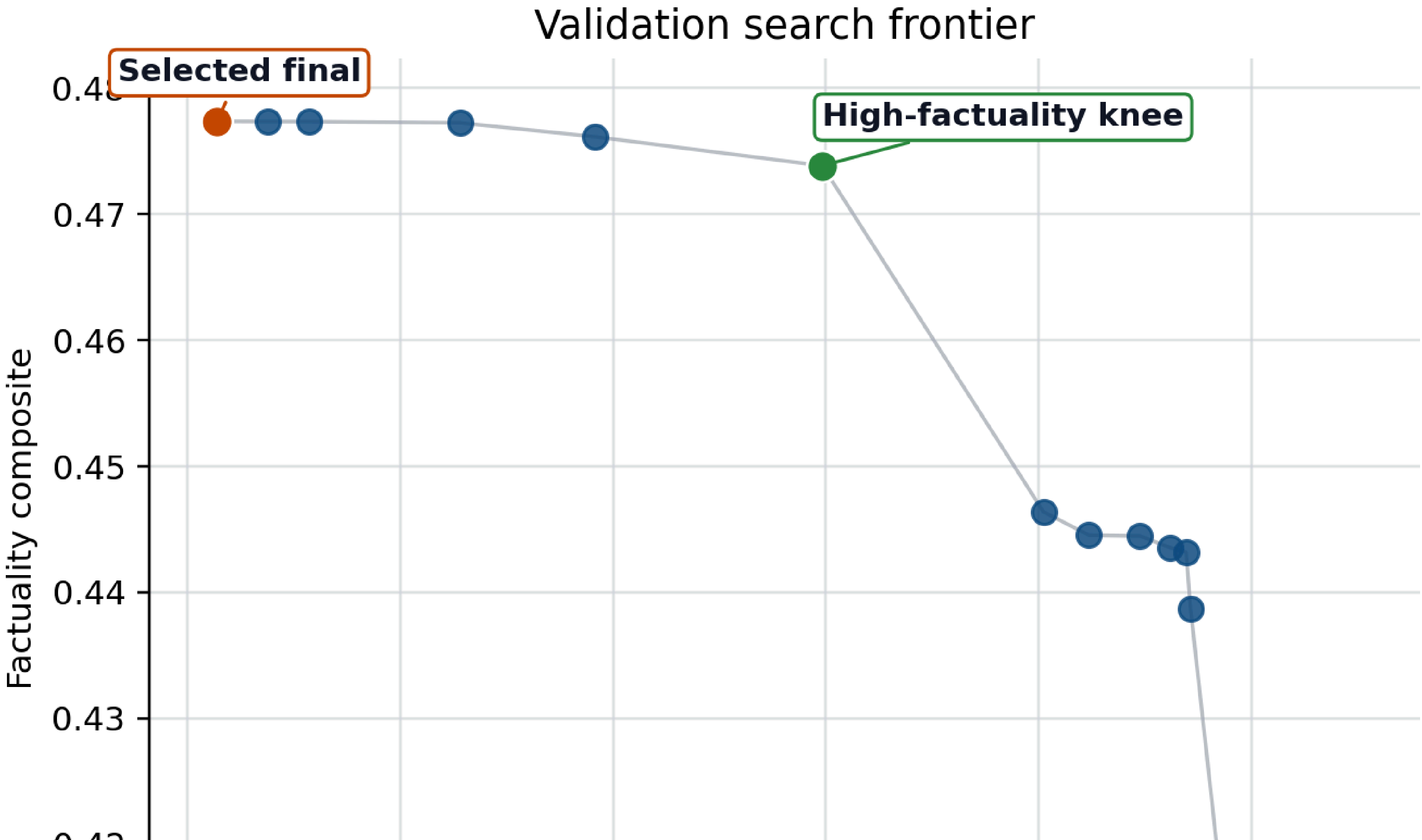
Method and Pipeline

- Fine-tune `facebook/bart-large-xsum` on the bounded train split.
- Generate beam candidates with sizes `4`, `8`, and `16`.
- Rerank with likelihood, SummaC-style support, FactCC-style consistency, and entity support.
- Select the operating point on validation search, then evaluate once on the test split.

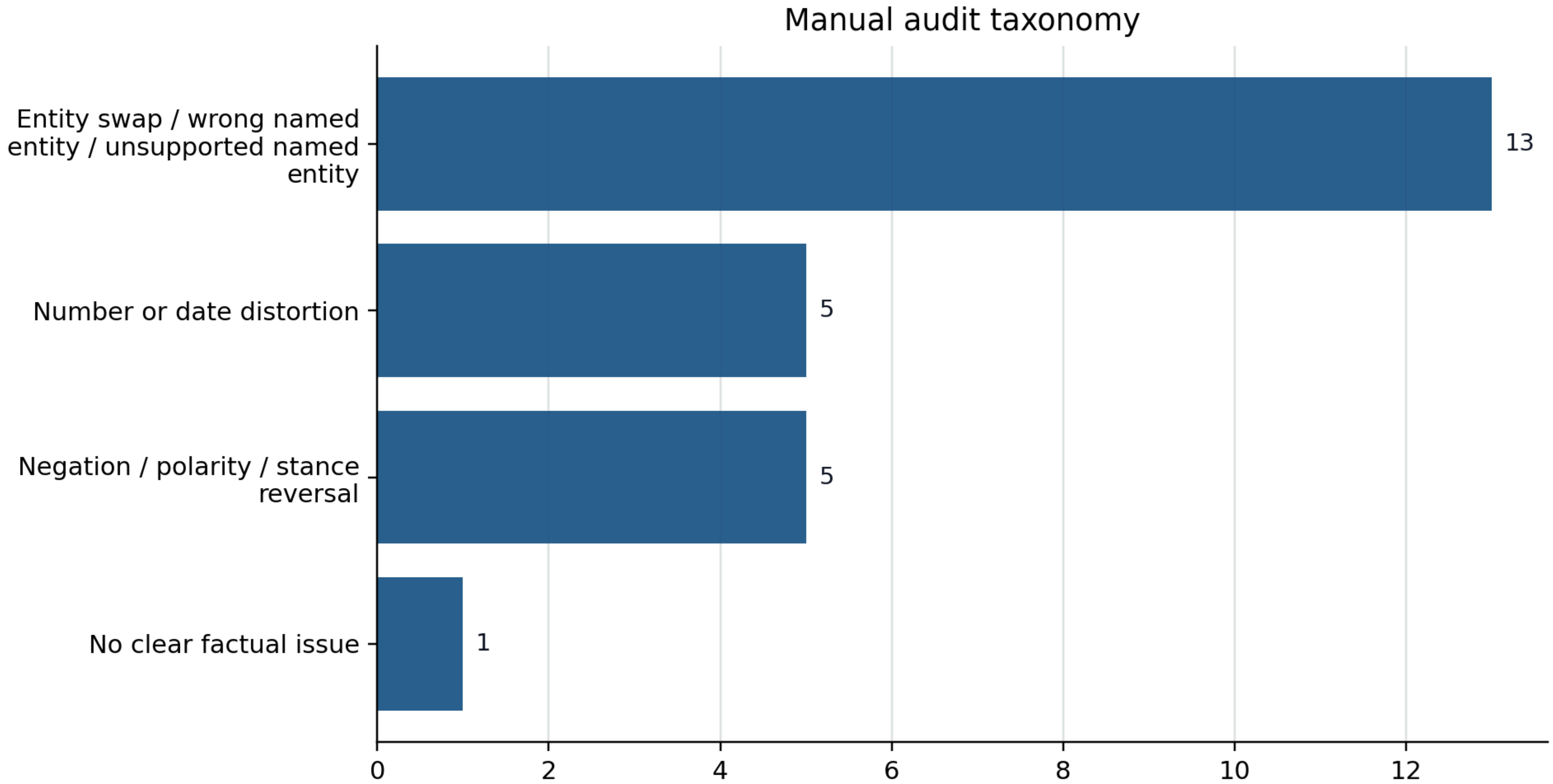
Pipeline Diagram



Search and Refinement



Qualitative Analysis and Limits



Takeaways

- Simple reranking signals materially improve factuality on this bounded XSum run.
- Entity support is a useful small refinement, not a silver bullet.
- The strongest remaining gap is entity-level hallucination when all candidates drift the same way.
- The next strongest improvements would likely come from better candidate diversity, stronger entity constraints, and broader external evaluation.